
Food recipe provenance

1 PROBLEM DESCRIPTION

The dataset consists of information from **4934 food recipes**. For **3200** of these recipes you know the **cuisine they belong to**: **Italian** ($y = 2$) or **American** ($y = 1$), and you have additional **input variables describing the ingredients of the recipe**. For the other **1734** recipes, you have only information on the inputs and not on the cuisine provenance.

Your goal is to predict y for the held-out **1734** recipes.

There are **40 input variables** measuring the composition of each recipe with respect to 40 main ingredients [e.g., **salt**, **sugar**, **water**, ..., **pepper**], where the generic input $x_{i,j}$, measures the relevance of ingredient j in recipe i , through the TERM FREQUENCY–INVERSE DOCUMENT FREQUENCY score. More specifically:

$$x_{i,j} = \text{TF}_{i,j} \times \text{IDF}_j$$

where $\text{TF}_{i,j}$ measures how often ingredient j appears in recipe i , normalized by the total number of times ingredients are mentioned in that recipe, while IDF_j quantifies how unique ingredient j is across all recipes in the dataset (in particular $\text{IDF}_j = \log(4934/n_j)$, where n_j is the number of recipes containing ingredient j).